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Systems and methods for managing the heating of thermal fluids

Abstract

A system for operating at least a pair of pumps using a driver. The driver can be coupled to a first portion of a first drive shaft that is not positioned in within an interior area of a thermal fluid tank of a first thermal circuit. A magnetic coupling can couple the first portion to a second portion of the first drive shaft that is within the thermal fluid tank. Rotational displacement of the first portion of the first drive shaft can be translated, via the magnetic coupling, to the second portion of the first drive shaft to drive a first fluid pump. The first portion of the first drive shaft can also be coupled to a second drive shaft by a clutch, the second drive shaft being used to operate a second fluid pump that provides a force for circulating a second thermal fluid of a second thermal circuit.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11739756	12/2022	Whiteman	N/A	N/A
11958361	12/2023	Smith	N/A	B60K 6/405
2013/0029193	12/2012	Dyer et al.	N/A	N/A
2014/0338376	12/2013	Carpenter et al.	N/A	N/A
2018/0229616	12/2017	Rhodes et al.	N/A	N/A
2022/0170467	12/2021	Whiteman	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102019131731	12/2020	DE	N/A
102020206146	12/2020	DE	N/A
102020204675	12/2021	DE	N/A
4215411	12/2022	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report and Written Opinion issued in European Patent Application No. 24192451.3 dated Feb. 12, 2025, in 05 pages. cited by applicant

Extended European Search Report and Written Opinion issued in European Patent Application No. 24192665.8 dated Feb. 21, 2025, in 11 pages. cited by applicant

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Background/Summary**FIELD OF THE DISCLOSURE**

(1) The present disclosure generally relates to management of thermal fluids used with batteries for electric vehicles, and, more particularly, to systems and methods for managing the warming of a battery thermal fluid and a cab of an associated vehicle.

BACKGROUND

(2) Hybrid electric engines typically operate using a combination of a fuel and electrical energy as sources of power for engine operation. Moreover, hybrid engines can combine features of an internal combustion engine and an electrically powered engine in generating a force, such as a force used with at least propulsion of an associated vehicle. Thus, power for hybrid engines can be

generated by combustion of a fuel, including, but not limited to, petroleum and diesel fuel, as well as by use of power stored in one or more batteries, including battery packs.

(3) Whether utilized with hybrid electric engines or electric engines, operation of associated battery systems, and, moreover, the battery(ies) of the battery system, can generate heat. If not controlled, such generated heat can be detrimental to the operation or efficiency of at least the battery. Often, the warming of batteries of electric or hybrid vehicles involves the circulation of an electrically non-conductive dielectric thermal liquid, such as, for example, mineral oil, and which is not air or another gas. However, the viscosity of such dielectric thermal fluids can increase at least at certain lower, or cold temperatures. In such instances, compared to instances at which the dielectric thermal fluid is at elevated temperatures, the relatively high viscous dielectric thermal fluid may not easily flow through the fluid circuit, which can be detrimental to the operation or efficiency of the fluid circuit. Further, instances in which the viscosity of the thermal fluid is relatively high can potentially create issues for the associated components of the fluid circuit, including premature wear of the pump(s) that generate a force to circulate the thermal fluid through the fluid circuit.

(4) Accordingly, thermal management systems for battery systems used with electric engines, including hybrid electric engines, remains an area of interest.

SUMMARY

(5) The present disclosure may comprise one or more of the following features and combinations thereof.

(6) In one embodiment of the present disclosure, a thermal management system is provided that can include a first fluid pump and a second fluid pump, the first fluid pump being positioned for at least partial submersion in a first thermal fluid in a thermal fluid tank, the second fluid pump positioned for at least partial submersion in a second thermal fluid in a first pressurized reservoir of a second thermal circuit. The system can also include a first drive shaft that can be coupled to a driver and the first fluid pump, the first fluid pump being operated by a rotational displacement of the first drive shaft by the driver. The first fluid pump can be configured to provide a first force to circulate the first thermal fluid out through the first fluid circuit. A second portion of the first drive shaft can be positioned in an interior area of the first thermal fluid tank, and a first portion of the first drive shaft can be positioned outside of the thermal fluid tank and not directly physically attached to the second portion of the first drive shaft. The system can also include a second drive shaft that can be coupled to the second fluid pump and coupled to the first portion of the first drive shaft by a clutch. The second fluid pump can be operated by a rotational displacement of the second drive shaft to provide a second force to circulate a second thermal fluid at least through a heat exchanger of a second fluid circuit that can be positioned in the thermal fluid tank. Additionally, the system can include a magnetic coupling having a first portion and a second portion, the first portion of the magnetic coupling that can be attached to the first portion of the first drive shaft and positioned outside of the thermal fluid tank. The second portion of the magnetic coupling can be coupled to the second portion of the first drive shaft and positioned in the interior area of the thermal fluid tank. The magnetic coupling transfer a force between the first and second portions of the magnetic coupling to rotatably displace the second portion of the magnetic coupling, and thereby rotate the second portion of the first drive shaft, with the rotational displacement of the first portion of the magnetic coupling as the first portion of the first drive shaft is rotably displaced by operation of the driver. Additionally, the clutch can be selectively displaceable from an engaged position at which the rotational displacement of the first portion of the first drive shaft is transmitted by the clutch to facilitate the rotational displacement of the second drive shaft, and a disengaged position at which the rotational displacement of the first portion of the first drive shaft is not transmitted by the clutch to facilitate the rotational displacement of the second drive shaft.

(7) In another embodiment, a method is provided for operating a thermal management system. The method can include determining if a temperature is below a predetermined temperature and activating, in response to determining the temperature is below the predetermined temperature, a

magnetic clutch to magnetically couple a first portion of the magnetic clutch to a second portion of the magnetic clutch. The first portion of the magnetic clutch can be coupled to a first portion of a first drive shaft that is positioned outside of a thermal fluid tank, and the second portion of the magnetic clutch can be coupled to a second portion of a first drive shaft that is positioned in an interior area of the thermal fluid tank, the second portion of the first drive shaft being coupled to a first fluid pump. A driver can be activated to provide a drive force to rotatably displace the first portion of the first drive shaft, the rotational displacement of the first portion of the first drive shaft being transmitted via the activated magnetic clutch to facilitate rotational displacement of the second portion of first drive shaft. Additionally, an operation of the first fluid pump can be driven by rotational displacement of the second portion of first drive shaft. A clutch that is coupled to the first portion of the first drive shaft and a second drive shaft can be engaged, the clutch being configured to transmit a portion of the drive force to rotatably displaced the second drive shaft. Further, an operation of a second fluid pump of a first pressurized reservoir can be driven by rotational displacement of the second drive shaft. The method can also include circulating, via the operation of the first fluid pump, at least a portion of a first thermal fluid out of the thermal fluid tank and along at least a portion of a first fluid circuit, and drawing, via the operation of the second fluid pump, a second thermal fluid into a heat exchanger in the thermal fluid tank. Additionally, the method can include determining if the temperature has increased to at least satisfy the predetermined temperature, and deactivating, in response to the temperature satisfied the predetermined temperature, the operation of the second fluid pump by disengaging the clutch.

(8) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure contained herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.
- (2) FIG. 1 illustrates an exemplary embodiment of a thermal management system according to an illustrated embodiment of the subject application in which a second thermal fluid is heated via operation of an electric heater, and in which heat is transferred from the heated second thermal fluid to a first, battery thermal fluid.
- (3) FIG. 2 illustrates the exemplary embodiment shown in FIG. 1, but in which the first, battery thermal fluid bypasses the battery and instead flows along a bypass flow path.
- (4) FIG. 3 illustrates the exemplary embodiment shown in FIG. 1, but in which a third thermal fluid of an external system is utilized to provide heat for at least the first, battery thermal fluid.
- (5) FIG. 4A illustrates an exemplary embodiment of a selectively activated electromagnetic clutch for the thermal management system in an engaged, or closed, condition such that both a first fluid pump and a second fluid pump of the thermal management system are operated by a common driver.
- (6) FIG. 4B illustrates the electromagnetic clutch shown in FIG. 4A but in a disengaged, or open, condition such that the first fluid pump, and not the second fluid pump, is operated.
- (7) Corresponding reference numerals are used to indicate corresponding parts throughout the several views.

DETAILED DESCRIPTION

- (8) While the concepts of the present disclosure are susceptible to various modifications and

alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will be described herein in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

(9) References in the specification to “one embodiment,” “an embodiment,” “an illustrative embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may or may not necessarily include that particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. Additionally, it should be appreciated that items included in a list in the form of “at least one A, B, and C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C). Similarly, items listed in the form of “at least one of A, B, or C” can mean (A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).

(10) In the drawings, some structural or method features may be shown in specific arrangements and/or orderings. However, it should be appreciated that such specific arrangements and/or orderings may not be required. Rather, in some embodiments, such features may be arranged in a different manner and/or order than shown in the illustrative figures. Additionally, the inclusion of a structural or method feature in a particular figure is not meant to imply that such feature is required in all embodiments and, in some embodiments, may not be included or may be combined with other features.

(11) A number of features described below may be illustrated in the drawings in phantom. Depiction of certain features in phantom is intended to convey that those features may be hidden or present in one or more embodiments, while not necessarily present in other embodiments. Additionally, in the one or more embodiments in which those features may be present, illustration of the features in phantom is intended to convey that the features may have location(s) and/or position(s) different from the locations(s) and/or position(s) shown.

(12) Embodiments described herein provide systems and methods for an efficient way for warming an operator cab of a vehicle, as well as warming the battery (ies) and associated thermal fluid of an electric vehicle, including, but not limited to, an electric hybrid vehicle.

(13) Compared to heat exchange between thermal fluids, an electric heater can be more effective for heating at least certain types of thermal fluids used in warming circuits, including for example, ethylene glycol and water, as well as combinations thereof, collectively referred to as WEG. Thus, as discussed below, embodiments of the subject disclosure provide a thermal management system having a fluid circuit in a primary system that can heat WEG via use of an electric heater. Such heated WEG can be utilized in connection with heating a thermal fluid that is supplied to one or more batteries, as well as in connection with heating an operator cab of the associated vehicle.

(14) As electric vehicles can be used in a variety of ambient temperature conditions, operation of the electric heater may not always be needed. Thus, to reduce energy consumption, as well as eliminate use of individual drivers for each pump of the thermal management system, embodiments discussed herein can utilize a single driver, double pump arrangement, wherein a common driver is used to provide a force for the operation of a first fluid pump that is used to circulate thermal fluid to the battery, and a second fluid pump used for circulating the WEG. With such an arrangement, a clutch is used to selectively activate, and deactivate, the second fluid pump. Moreover, the clutch can be in a disengaged or open state in warm weather conditions such that the second fluid pump is not operated, which can thereby reduce power usage. Conversely, during colder ambient temperatures when heat may be needed for the thermal fluid circulated to the battery, or for heating the operator cab, the clutch can be in an engaged or closed state such that both the first and second

fluid pumps are driven by the common driver.

(15) Additionally, when the onboard battery is unable to provide electrical power for the electric heater, the thermal management system discussed herein includes a provision for heating thermal fluid that is circulated to the battery, and thus warm the battery, using heated WEG from an external system. Such hot WEG from the external system can be provided to a primary system of the thermal management system in a variety of manners, including, for example, via coupling the primary system to an external system via use of external couplings.

(16) When the electric vehicle is expected to sit outside in cold weather without charging, the electric vehicle can also benefit from pumping/purging the thermal fluid from the battery and storing the purged thermal fluid in an insulated thermal fluid tank overnight. This feature can allow for more efficient warming of the thermal fluid overnight, without losing heat in the batteries and plumbing.

(17) FIG. 1 illustrates an exemplary embodiment of a thermal management system **100** according to an illustrated embodiment of the subject application. In the illustrated embodiment, the thermal management system **100** includes a primary system **102** and an external system **104**. The primary system **102** can be associated with an electric vehicle **106**, including, but not limited to, a hybrid electric vehicle. Thus, the primary system **102** can be part of an electric vehicle **106** having one or more batteries **108** that are utilized to provide electrical power for at least operation of a prime mover, including engine, of the electric vehicle **106**. Further, the primary system **102** can be utilized with a variety of different types of electric vehicles, including, but not limited to, agricultural vehicles, such as, but not limited to, harvesters and windrowers, among other agricultural vehicles. According to certain embodiments, the electric vehicle **106** includes an operator cab **110** in which an operator of the electric vehicle **106** can be positioned, including either, or both, seated or stand, while operating the electric vehicle **106**.

(18) The external system **104** can be part of another vehicle, equipment, or device (referred to herein collectively as an external device **112**), that is not part of the electric vehicle **106** having the primary system **102**. Additionally, the external system **104** can be part of, or coupled to, a coolant or heating system of the external device **112** to which the external system **104** belongs. Thus, for example, according to embodiments in which the external device **112** has an engine, among other prime movers, and the external system **104** can be utilized as part of a cooling system for that engine, and well as selectively used as a fluid circuit with the primary system **102** for the below-discussed thermal management system **100**. Thus, for example, the external system **104** can include, or be coupled to, a heat exchanger **114**, such as a radiator, that is used in connection with cooling a temperature of the engine of the external device **112**. In such situations, the external system **104** can contain a thermal fluid that may directly flow to the engine of the external device **112**, wherein a temperature of the thermal fluid of the external system **104** is elevated.

Alternatively, the external system **104** can be configured for the associated thermal fluid to absorb at least a portion of heat entrained in another thermal fluid, such as, for example, at the heat exchanger **114** of the external system **104**.

(19) While the foregoing example discusses heating of the thermal fluid of the external system **104** in connection with an operation of an associated engine, the thermal fluid of the external system **104** can be heated via operation of a variety of other components of the external device **112**, including, but not limited to, one or more batteries, inverters, or converters, among other electrical and non-electrical components of the external device **112**. Additionally, while the primary system **102** is discussed herein as being utilized with an electrical vehicle, the external device **112** may, or may not, be an electrical vehicle. For example, according to certain embodiments, the external device **112** can be a personal or commercial vehicle that utilizes an internal combustion.

(20) The exemplary thermal management system **100** includes a plurality of fluid circuits **116a-c**, with at least one fluid circuit **116c** extending between, or be part of, both the primary system **102** and the external system **104**. The thermal fluid within each illustrated fluid circuit **116a-c** can be

transported within the associated fluid circuit **116a-c**, as well as between, and, optionally, within, the associated components of the fluid circuit **116a-c**, through a variety of types of conduits, tubes, hoses, or pipes, as well as combinations thereof, and the associated fittings, valves, and ports, collectively generally referred to herein as a conduit **118**. As discussed below, certain fluid circuits **116a** are fluidly isolated from other fluid circuits **116b**, **116c** so as to at least prevent mixing of the thermal fluids contained therein, while other circuits **116b**, **116c** can be adapted to accommodate at least potential partial exchange or mixing of similar types of thermal fluids.

(21) The disclosures contained herein discuss the operation of the thermal management system **100** in the context of either or both warming a first thermal fluid for a battery **108**, or warming an operator cab **110** of the electric vehicle **106**. However, according to certain embodiments, the fluid circuits **116a-c**, or thermal fluids contained therein, can also be utilized in connection with other aspects of thermal management of at least the battery **108**, associated electric vehicle **106**, or another, external device **112**. For example, the below disclosure discusses operation of the thermal management system **100** in connection with elevating a temperature of a first thermal fluid that circulates through the battery **108**, which can reduce the viscosity of the first thermal fluid. However, at other times, the first thermal fluid circulated through one of the more the below-discussed fluid circuits **116a** can also be used for other purposes, including for example, in connection with maintaining temperatures of the battery **108** or an associated engine, prime mover, or electrical components below certain elevated temperatures.

(22) According to the illustrated embodiment, the thermal management system **100** has a first fluid circuit that can selectively accommodate circulation of a first thermal fluid along either a primary flow path **116a-1** or a bypass flow path **116a-2** of the first fluid circuit **116a**. As shown in at least FIG. **1**, the first fluid circuit **116a** can include one or more batteries **108**, an inlet control valve **120**, and outlet control valve **122**, a first fluid pump **124**, a thermal fluid tank **126**, and a bypass conduit **118a**. As seen in at least FIG. **1**, in the illustrated exemplary embodiment, the bypass conduit **118a** can extend between the inlet and outlet control valves **120**, **122**.

(23) The primary flow path **116a-1** of the first fluid circuit **116a** is configured to deliver the first thermal fluid to at least the battery **108**. Conversely, the bypass flow path **116a-2** is configured to bypass the battery **108** such that the first thermal fluid being circulated through the bypass flow path **116a-2** does not circulate through the battery **108**. Whether the first thermal fluid flows through the primary flow path **116a-1** or the bypass flow path **116a-2** can be controlled in a variety of manners, including, for example, via the positioning or operation of the inlet and outlet control valves **120**, **122**. For example, according to certain embodiments, when the inlet and outlet control valves **120**, **122** are in a first position, the flow path of the first thermal circuit **116a** is open in a manner that allows the first thermal fluid to flow along the primary flow path **116a-1**, and thereby to and through the battery **108**. Conversely, when the inlet and outlet control valves **120**, **122** are at a second position, the flow path of the first thermal circuit **116a** is open in a manner that redirects the first thermal fluid to flow through the bypass conduit **118a** that is positioned between the inlet and outlet control valves **120**, **122**, and thereby not be circulated through the battery **108**. As discussed below, according to certain embodiments, the bypass flow path **116a-2** can be utilized in at least an attempt to decrease the time before a temperature of the first thermal fluid is elevated to a particular level.

(24) As the first thermal fluid is, via the primary flow path **116a-1** of the first fluid circuit **116a**, circulated to at least the battery **108**, the first thermal fluid is a dielectric thermal liquid, and is not air or another gas. Moreover, the first thermal fluid is a liquid that can be utilized to remove heat from the battery **108**, including, for example, from another thermal fluid within the battery **108**, while also being electrically non-conductive. By being electrically non-conductive, the first thermal fluid may not cause electrical short circuiting within the battery **108**, or across at least the first fluid circuit **116a** or the associated components of the first fluid circuit **116a**. Thus, the first thermal fluid may instead generally be an electric insulator. The first thermal fluid can, at least when at elevated

temperatures, also have a relatively high thermal conductivity and a relatively low viscosity, which can assist in the first thermal fluid being circulated about the first fluid circuit **116a**. However, the first thermal fluid can have a relatively high viscosity when the first thermal fluid is at low or cold temperatures. According to certain embodiments, the first thermal fluid can be mineral oil, as well as various commercially available dielectric thermal fluids, among other dielectric liquids.

(25) As seen in at least FIGS. **1**, **4A**, and **4B**, the first fluid pump **124** is coupled to a driver **128** by a first drive shaft **130**. A variety of devices can be utilized for the driver **128**, including, for example, an electric motor or actuator. The force generated by the driver **128** can be transmitted to the first fluid pump **124** via at least the first drive shaft **130**. Operation of the first fluid pump **124** via at least a portion of the force provided by the driver **128** can provide a force for circulating the first thermal fluid through the first fluid circuit **116a**, including the primary flow path **116a-1** and the bypass flow path **116a-2**. As seen in FIG. **1**, according to certain embodiments, the first fluid pump **124** can be submerged in first thermal fluid within the thermal fluid tank **126**.

(26) An interior region **132** of the thermal fluid tank **126** can contain a collection of the first thermal fluid. Further, according to certain embodiments, the thermal fluid tank **126** can be a thermally insulated and vented tank. Additionally, the thermal fluid tank **126** can have an upper portion **134** and a lower portion **136**, the upper portion **134** being positioned above, or at a higher vertical position than, the lower portion **136**. As heat typically rises, the first thermal fluid at or around the upper portion of the thermal fluid tank **126** can have a temperature that is higher than the temperature of the first thermal fluid that is located at or around the lower portion **136** of the thermal fluid tank **126**. Thus, as thermal gradients may develop in the thermal fluid tank **126**, a standpipe **138** can be used by, or in connection with, the first fluid pump **124** in at least an attempt to pull the hotter first thermal fluid from the top or upper portion **134** of the thermal fluid tank **126** and into the standpipe **138**. Thus, for example, the standpipe **138** can include an inlet **140** that is positioned at or around the upper portion **134** of the thermal fluid tank **126**. Moreover, the inlet **140** of the standpipe **138** can be positioned so that the first thermal fluid at or around the inlet **140** is drawn, or flows, into the standpipe **138**. Accordingly, to the extent a temperature difference exists between the first thermal fluid in the thermal fluid tank **126**, the standpipe **138** can assist with the hotter first thermal fluid located at or around the upper portion **134**, rather than the cooler first thermal fluid that may be located around the lower portion **136**, of the thermal fluid tank **126** being discharged from the thermal fluid tank **126** and delivered to the battery **108**.

(27) The primary system **102** can also include a second fluid circuit **116b** that circulates a second thermal fluid that is different than the first thermal fluid. More specifically, compared to the first thermal fluid, the second fluid circuit **116b** can circulate an electrically conductive second thermal fluid, including, for example, either, or a mixture of, water or glycol, among other conductive fluids. Additionally, the thermal management system **100** can include a second fluid pump **142** that can provide a force to circulate at least the second thermal fluid through the second fluid circuit **116b**. According to certain embodiments, the second fluid pump **142** can be positioned to withdraw a heated second thermal fluid from a pressurized reservoir **144** of the second fluid circuit **116b**. As discussed below, the second fluid pump **142** can be selectively operated using the same driver **128** that is used for the operation of the first fluid pump **124**, and can be submerged in second thermal fluid within the pressurized reservoir **144**.

(28) As seen in at least FIG. **1**, the second fluid circuit **116b** can include an electric heater **146** that can be selectively powered by electrical power provided by the battery **108**. According to certain embodiments, the second thermal fluid can flow, via a force provided by the second fluid pump **142**, from the electric heater **146** to a first heat exchanger **148** that can be submerged within first thermal fluid within the thermal fluid tank **126**. Alternatively, according to certain embodiments, second thermal fluid that is heated by the electric heater **146** can flow to a pressurized reservoir **144**, which can include a pressure relief cap **150** (FIG. **4A**), before being circulated to the first heat exchanger **148**. The second thermal fluid within the second fluid circuit **116b**, including within the

pressurized reservoir **144**, can be maintained at a variety of pressures, including, but not limited to, pressures between around 10 pounds-per-square inch (PSI) to around 20 PSI, among other pressures or pressure ranges. According to such an embodiment, heated second thermal fluid can be drawn from the pressurized reservoir **144** and into the second fluid pump **142** before being delivered to the first heat exchanger **148**. The first heat exchanger **148** within the thermal fluid tank **126** can be utilized to transfer at least a portion of the heat entrained in the second thermal fluid to at least a portion of the first thermal fluid that is in the thermal fluid tank **126**. A variety of different types of heat exchangers can be utilized for the first heat exchanger **148**, as well as other heat exchangers **114**, **152** discussed herein, including, but not limited to, brazed plate heat exchangers, as well as one or more tubes that pass through the interior region **132**, among types or configurations of heat exchangers.

(29) According to certain embodiments, the second thermal fluid that is outputted from the thermal fluid tank **126** can be delivered to the operator cab **110** such that heat within the second thermal fluid can be utilized to heat the operator cab **110**. Thus, according to certain embodiments, the operator cab **110** can include a second heat exchanger **152** that can be selectively used to transfer heat from the second thermal fluid to air, or other gas, that can be circulated within the operator cab **110**. The operator cab **110** can have variety of types of components and devices to control if, and the extent, air heated by the second heat exchanger **152** is released or circulated within the operator cab **110**, including, or more convection fans **154**, as well as associated vents.

(30) A third fluid circuit **116c** can utilize a third thermal fluid that can be the same type of thermal fluid as the second thermal fluid. However, unlike the second fluid circuit **116b** in which the primary source for the second thermal fluid can be the associated electric vehicle **106**, the primary source of the third thermal fluid used for the third fluid circuit **116c** can be the external device **112**. Further, as the second and third fluid circuits **116b**, **116c** can be utilized at least some of the same components and conduits **118**, at least a portion of the second thermal fluid provided by the electric vehicle **106** can intermix with at least a portion of the third thermal fluid provided by the external device **112**, and vice versa. Moreover, according to the illustrated embodiment, a first portion **116c-1** of the third fluid circuit can reside with the external system **104** at the external device **112**, and a second portion **116c-2** can reside with the primary system **102**, and, moreover, a portion of the second fluid circuit **116b**, at the electric vehicle **106**. Thus, according to certain embodiments, the second fluid circuit **116b** may not be fluidly isolated from the third fluid circuit **116c**.

(31) As seen in at least FIG. **1**, the second and third fluid circuits **116b**, **116c** can both utilize the first heat exchanger **148** in the thermal fluid tank **126**, and the second heat exchanger **152** for the operator cab **110**, as well as the associated conduits **118**. Accordingly, similar to the second fluid circuit **116b**, the third fluid circuit **116c** can pass through the operator cab **110** such that the at least partially heated third thermal fluid flowing through the third fluid circuit **116c** can also be utilized to heat the operator cab **110**. Thus, according to certain embodiments, the third thermal fluid may not be isolated from the second thermal fluid. However, the second and third fluid circuits **116b**, **116c** can be fluidly isolated from the first fluid circuits **116a** such that the first thermal fluid does not mix with the second and third thermal fluids.

(32) As previously discussed, the external system **104** can include a heat exchanger **114** that can be utilized to heat the third thermal fluid. Additionally, according to certain embodiments, the third fluid circuit **116c** can also include a pressurized reservoir **156** that can include a supply of the third thermal fluid. According to such embodiments, the pressurized reservoir **156**, which can include a pressure relief cap **150** (FIG. **4A**), can be configured to store heated third thermal fluid. Moreover, the reservoir **156** can be configured to store the third thermal fluid at elevated pressures such that the reservoir **156** is a pressurized reservoir **156** that maintains the heated third thermal fluid at an elevated pressure, including, for example, at pressures in the range of around 10 PSI to around 20 PSI, among other pressure levels or ranges.

(33) As the third thermal fluid can be heated by the external device **112**, the third fluid circuit may

bypass, or otherwise not include, the electric heater **146**. Thus, the second and third fluid circuits **116b**, **116c** can include a circuit control valve **158** that controls whether the second thermal fluid of the second fluid circuit **116b** that has been heated by the electric heater **146**, or the third thermal fluid of the third fluid circuit **116c** that has been heated by the heat exchanger **114** of the external device **112**, is to be delivered to the first heat exchanger **148** within the thermal fluid tank **126**. (34) Additionally, the third fluid circuit **116c** can include one or more inlet ports **160a**, **160b** and outlet ports **162a**, **162b** that can each be selectively coupled to a conduit **118b**, **118c**, such as, for example, a hose, tube, or pipe, such the portion of the third fluid circuit **116c** residing with the external system **104** can be selectively connected, and disconnected, from the portion of the third fluid circuit residing with the primary system **102**. For example, according to certain embodiments, the inlet and outlet ports **160a-b**, **162a-b** can each comprise a quick connect hose connector or fitting, among other fittings and connectors.

(35) Referencing FIGS. **4A** and **4B**, the first and second fluid pumps **124**, **142** can be in a double pump drive arrangement. Moreover, a single driver **128** can be utilized to drive both the first and second fluid pumps **124**, **142**. For example, as shown in FIGS. **4A** and **4B**, the driver **128** can be coupled to the first fluid pump **124** such that operation of the driver **128** results in associated operation of the first fluid pump **124**. Moreover, the first fluid pump **124** can be coupled to the driver **128** in a manner in which the first fluid pump **124** is operated each time that the driver **128** is operated. Thus, when the driver **128** is operated, the driver **128** facilitates rotational displacement of a first drive shaft **130**, and thus associated operation of the first fluid pump **124**.

(36) Unlike the operation of the first fluid pump **124**, the second fluid pump **142** can be coupled to the driver **128** in a manner that accommodates selective operation of the second fluid pump **142** while the driver **128** and first fluid pump **124** are operated. For example, according to certain embodiments, a clutch **164**, including, for example, an electromagnetic clutch, can be utilized to selectively accommodate rotational displacement of a second drive shaft **166** that is coupled to, and used to operate, the second fluid pump. According to the example embodiment shown in FIGS. **4A** and **4B**, the clutch **164** can include a rotor **168** that is coupled to the first drive shaft **130** so that the rotor **168** rotates with the rotation of the first drive shaft **130**. The clutch **164** can also include one or more coils **170** that are positioned about, or around, the rotor **168**, and which are electrically coupled to at least one battery **108**, among other sources of electrical power. A supply of electrical power to the coil **170** can result in the generation of a magnetic field at or around the coil **170**.

(37) The clutch **164** can further include one or more clutch bodies **172** that are constructed of a material that can either, or both, be attracted or repelled by the magnetic field(s) generated via the supply of electrical power to the coil **170**. According to certain embodiments, the clutch body **172**, which can be one or more magnets, can be attached to a biasing element **174** that can bias the clutch body **172** to one of a first position (FIG. **4A**) and a second position (FIG. **4B**). Additionally, or alternatively, the clutch **164** can be configured to accommodate at least displacement of the clutch body **172** in response to a magnetic field provided by, or removed, via the operation coil(s) **170**. Thus, for example, according to certain embodiments, the magnetic field generated by use of the coil **170** can have a strength to attract or repel the clutch body **172** with a force sufficient to overcome the biasing force of the biasing element **174**, thereby displacing the clutch body **172**. Further, the clutch body **172** can be coupled to, or part of, the rotor **168** such that the clutch body **172** is rotatably displaced with the rotor **168** as the driver **128** provides a force that rotates the first drive shaft **130**.

(38) The clutch **164** can also include an armature **176** that is coupled to the second drive shaft **166** such that rotation of the armature **176** can facilitate rotation of the second drive shaft **166**. According to the illustrated embodiment, the armature **176** can include one or more armature bodies **178** that are each configured to be selectively engaged and disengaged with the clutch body **172**. For at least purposes of illustration, the armature bodies **178** comprise one or more recesses **180**, each recess **180** being sized to receive placement of a clutch body **172** when the clutch body

172 is at a first position (FIG. **4A**). According to the illustrated embodiment, the clutch body **172** can be at the first position via a force provided by the biasing element **174**. By being within a recess of the armature **176**, the clutch body **172** can engage, or abut against, a surface or wall of the armature **176**. Using such engagement between clutch body **172** and armature **176**, as the clutch body **172** and rotor **168** rotate with the rotation of the first drive shaft **130**, the clutch body **172** can transmit a force to the armature **176** that facilitates rotational displacement of the armature **176** and thus the second drive shaft **166**. Such rotational displacement of the second drive shaft **166** can thus provide a force utilized to activate, and operate, the second fluid pump **142**.

(39) Further, according to the illustrated example, the clutch body **172** can be displaced from the first, engaged or closed position (FIG. **4A**) to the second, disengaged or open position (FIG. **4B**). Moreover, in this example, the magnetic field generated by a supply of electrical power to the coil **170** can generate a magnetic force that overcomes the biasing force of the biasing element **174**, and pulls or displaces the clutch body **172** away from the armature **176**. In such a situation, the clutch body **172** can be moved to a location at which the clutch body **172** does not, or cannot, engage an armature body **178**. With the clutch body **172** disengaged from a corresponding armature body **178**, rotational displacement of the clutch body and rotor **168** is not transferred to the armature **176**. Thus, with the clutch body **172** disengaged from the armature **176**, force provided by driver **128** that rotates the first drive shaft **130** such that the first fluid pump **124** is operated, but does not rotate the second drive shaft **166**, thereby causing the second fluid pump **142** to be deactivated, or otherwise not operate.

(40) The above discussed configuration accommodates selective operation of the second fluid pump **142** while the first fluid pump **124** can generally be continuously operated via operation of the driver **128**. Such selective operation of the second fluid pump **142** can, according to certain embodiments, be generally limited to during specific low temperature ambient conditions. More specifically, the second fluid pump **142** can be disengaged, via operation of the clutch **164**, when heating of the first thermal fluid by the second thermal fluid is not needed. Such a one motor, dual pump configuration can provide benefits with respect to at least space savings in terms of the footprint of the motor/pump system, as well as provide savings associated with component costs, installation, and maintenance. Additionally, utilizing a single driver **128** can potentially improve the reliability of the system.

(41) The first and second fluid pumps **124**, **142** can also be utilized to maintain the associated first and second thermal fluids within certain predetermined pressure limits. According to certain embodiments, the magnetic coupling provided by the clutch **164** can be adapted to control, or limit, the torque exerted by the clutch **164** onto the armature **176**, and thus control the torque exerted by the second drive shaft **166** in operating the second fluid pump **142**. By limiting the torque, the operation of the second fluid pump **142** can be controlled in a manner in which the operation of the second fluid pump **142** does not exceed a pump pressure limit, including a pressure limit for the second thermal fluid and the third thermal fluid in the second and third fluid circuits **116b**, **116c**, respectively. Such limiting of the torque outputted by the clutch **164** can be achieved in a variety of manners, including, for example, via the number and spacing of clutch bodies **172**, and more specifically magnets. Further, configuring the torque transmitted by the clutch **164**, and thus used in the operation of the second fluid pump **142** so as to generally match a pump or thermal fluid pressure limit can allow for an elimination of a pressure relief valve for the second fluid circuit **116b**.

(42) While the foregoing discusses an exemplary clutch **164**, the clutch **164** described herein can include a variety of other components, including bearings, drive hubs, and flanges, among other components. Additionally, a range of other clutch configurations, including different electromagnetic clutch configurations, can also be utilized for the clutch **164** of the thermal management system **100** discussed and described herein.

(43) Additionally, according to other embodiments, rather than utilizing a common driver **128**, each

of the first and second fluid pumps **124**, **142** can be operated using a different, or separate driver **128**. According to such an embodiment, a first driver can be utilized to operate the first fluid pump **124**, and a second driver can be utilized to operate the second fluid pump **142**. Thus, with such embodiments, whether, or when, thermal fluid is to flow through the associated fluid circuit **116a-c** can be controlled by separate, or individual, activation and deactivation of the associated first and second drivers.

(44) As previously discussed, the first and second fluid pumps **124**, **142**, and at least portions, if not all, of the first and second drive shafts **130**, **166**, can be submerged in thermal fluid within the thermal fluid tank **126** and pressurized reservoir **144**, respectively. For example, as seen in at least FIGS. **4A** and **4B**, according to certain embodiments, a first portion **165a** of a magnetic coupling **165** can be coupled to a portion of the first drive shaft **130** that is outside of the interior region **132** of the thermal fluid tank **126**. Further, a second portion **165b** of the magnetic coupling **165** can be coupled to a portion of the first drive shaft **130** that is inside of the interior region **132** of the thermal fluid tank **126**. According to such an embodiment, the rotational force being used to rotate the portion of the first drive shaft **130** that is outside of the interior region **132** of the thermal fluid tank **126** can be transmitted to rotate the portion of the first drive shaft **130** that is inside of the interior region **132** via the magnetic coupling **165**. Thus, such an embodiment can accommodate rotation of the first drive shaft **130** within the interior region **132** so as to provide a force for operation of the first fluid pump **124** without the first drive shaft **130** passing through, or otherwise breaching, a sidewall of the thermal fluid tank **126**. Such a configuration can thereby eliminate the use of pump shaft seals. As pump shaft seals traditionally can be a potential area of weakness in the durability of pumps, eliminating pump shaft seals can improve the durability of either or both the first and second fluid pumps **124**, **142**.

(45) While the foregoing example is discussed and illustrated with respect to use of a magnetic coupling **165** with the first drive shaft **130** and the thermal fluid tank **126**, additionally, or alternatively, a similar magnetic coupling **165** can be utilized with the second drive shaft **166** and the pressurized reservoir **144** in connection with operation of the second fluid pump **142**. Additionally, as shown in FIGS. **1-4B**, according to certain embodiments, the clutch **164** and first portion **165a** of the magnetic coupling **165** can be housed in a drive housing **167** that is separate, and may be fluidly isolated, from the thermal fluid tank **126** and the pressurized reservoir **144**.

(46) The thermal management system **100** can also include a controller **182** having one or more processors **184** that can follow instructions, including control instructions, contained within, or are part of, one or more memory devices **186**, including, for example, a non-transitory machine readable medium. The exemplary embodiment shown in at least FIG. **1** illustrates the controller **182** as being separate from the primary system **102**, and instead being part of another portion of the electric vehicle **106**. However, according to other embodiments the primary system **102**, or other system of the electric vehicle **106**, can have a controller **182** that is dedicated to the operation of the primary system **102** alone or in conjunction with the external system **104**. As also shown in at least FIG. **1**, the controller **182** can be electrically coupled to the battery **108**, and moreover may be powered by operation of the battery **108**.

(47) The thermal management system **100** can also include one or more temperature sensors **188** that can be communicatively coupled to the controller **182**. According to the illustrated embodiment, at least one temperature sensor **188** is positioned within the thermal fluid tank **126**, and can provide information indicating, or used to determine, a temperature of the first thermal fluid contained within the thermal fluid tank **126**. Information provided to the controller **182** by the temperature sensor **188** can provide an indication to the controller **182**, including, for example, the processor **184**, of whether the second thermal fluid is to be circulated at least through the thermal fluid tank **126** to heat the first thermal fluid. Thus, using such information, the controller **182** can provide instructions that can result in the battery **108** providing electrical power to the coil(s) **170** of the clutch **164** in a manner that can engage the clutch body(ies) **172** to engage the armature **176**

so as to facilitate rotational displacement of the second drive shaft **166** and associated operation of the second fluid pump **142**. Further, to the extent the driver **128** does not continuously operate, the controller **182**, including, for example, the processor **184**, can generate instructions that can activate, as well as deactivate, operation of the driver **128**.

(48) The controller **182**, including, for example, the processor **184**, can operate a plurality of control valves **120**, **122**, **158**, **196** of the primary system **102**. For example, the controller **182** can be utilized to operate the inlet and outlet control valves **120**, **122** so as to control whether first thermal fluid flows along the primary flow path **116a-1** or bypass flow path **116a-2** of the primary fluid circuit **116**. Additionally, the controller **182**, including, for example, the processor **184**, can control the operation of the circuit control valve **158** so as to control whether second thermal fluid of the second fluid circuit **116b** or the third thermal fluid of the third fluid circuit **116c** flows to the thermal fluid tank **126**. As discussed below, such determination by the controller **182** can be based on a variety of factors including, for example, whether the external system **104** is fluidly coupled to the primary system **102**, and whether the controller **182** identifies the battery **108** as having sufficient electrical power available for operating the electric heater **146**.

(49) With respect to the third fluid circuit **116c**, the controller **182** can also determine whether the external system **104** is fluidly coupled to the primary system **102** in a variety of manners, including, for example, via use of a sensor **190**. For example, according to certain embodiments, the primary system **102** can include a pressure, flow or temperature sensor **188**, as well as a combination thereof, among other sensors. Although the sensor **190** shown in at least FIG. **1** is positioned between the inlet port **160a** of the primary system **102** and the circuit control valve **158**, the sensor **190** can be positioned along a variety of other locations about the third fluid circuit **116c**. Additionally, according to certain embodiments, the sensor **190** can detect either or both an inlet conduit **118b** or an outlet conduit **118c** being coupled to the associated inlet or outlet port **160a**, **162a** of the primary system **102** in a manner that can indicate that external system **104** is fluidly coupled to the primary system **102**.

(50) Referring again to FIG. **1**, FIG. **1** illustrates a scenario in which the primary flow path **116a-1** of the first fluid circuit **116a** is open, as indicated by solid line format for the associated conduits **118** and associated components. In this scenario, the bypass flow path **116a-2** of the first fluid circuit **116a** is closed, as indicated by the broken line format for the associated conduits **118** and components. Additionally, in the scenario illustrated in FIG. **1**, the second fluid circuit **116b** is open, while the third fluid circuit **116c** is closed, as also indicated by the solid and broken line formats, respectively, of the illustrated conduits **118** and associated components. Such a scenario can arise, for example, in situations in which the temperature of the first thermal fluid is to be elevated, and the battery **108** has sufficient power to operate the electric heater **146**. Additionally, such a scenario can arise when the external system **104** is not fluidly coupled to the primary system **102**, including, for example, when the inlet and outlet conduits **118b**, **118c** are not coupled to the associated inlet and outlet ports **160a-b**, **162a-b**.

(51) According to the scenario illustrated in FIG. **1**, the controller **182** can receive a signal from the temperature sensor **188** in the thermal fluid tank **126** indicating the temperature of the first thermal fluid is to be increased. In such a situation, to the extent not already at the first position, controller **182** can generate a signal to place the inlet and outlet control valves **120**, **122** both at the first position such that first thermal fluid flows through the battery **108**, and not through the bypass conduit **118a**. Further, to the extent necessary, the controller **182** can also generate a signal to facilitate the circuit control valve **158** being operated to place the circuit control valve **158** at a first position such that the second thermal fluid of the second fluid circuit **116b** can be circulate through the electric heater **146**, and the first heat exchanger **148** in the thermal fluid tank **126**. Further, the controller **182** can generate a signal to facilitate the clutch **164** being at the first, engaged position such that the rotational force provided by the driver **128** is used to rotate both the first and second drive shafts **130**, **166**, and thereby result in operation of the associated first and second fluid pumps

124, 142. For example, according to the exemplary illustrated embodiment, the controller **182** can generate a signal that accommodates the biasing force of the biasing element displacing the clutch body **172** into engagement with the armature **176**. The controller **182** can also generate a signal to facilitate the battery **108** providing power to operate the electric heater **146**, and, moreover, for the electric heater **146** to be operated in a manner that heats the second thermal fluid.

(52) Accordingly, with the scenario shown by FIG. **1**, with the second fluid pump being operated, second thermal fluid that has been heated by the electric heater **146** can be circulated to the first heat exchanger **148** in the thermal fluid tank **126**. Optionally, according to certain embodiments, after being heated, and before being circulated by the second fluid pump **142** to the first heat exchanger **148**, the heated second thermal fluid can be delivered to the pressurized reservoir **144**. Thus, according to such an embodiment, the heated second thermal fluid may be drawn from the pressurized reservoir **144** of the second fluid circuit **116** before being circulated to the first heat exchanger **148**.

(53) By passing through the first heat exchanger **148**, heat entrained in the second thermal fluid can be transferred to at least a portion of the first thermal fluid that is contained in the thermal fluid tank **126**. Upon exiting the thermal fluid tank **126**, the second thermal fluid, which is not intermixed with the first thermal fluid in the thermal fluid tank **126**, can still remain partially heated, or otherwise be at an elevated temperature. Thus, the second thermal fluid can be circulated from the thermal fluid tank **126** and be delivered to the second heat exchanger **152** in, or around, the operator cab **110**. At least a portion of the heat remaining within the second thermal fluid can then be transferred to at least a portion of an air or gas that is to be circulated in, and used to heat, the operator cab **110**. For example, as shown at least FIG. **1**, the controller **182** can be used to operate a fan **154** that can circulate air through or around the second heat exchanger **152** so as to heat the air that is to be circulated in the operator cab **110**. The at least partially cooled second thermal fluid can then flow from the second heat exchanger **152** and to the electric heater **146**, wherein the electric heater **146** can again reheat the second thermal fluid.

(54) The process of circulating the second thermal fluid through the second fluid circuit **116b** can continue until the temperature sensor **188** provides a signal to the controller **182** indicating that the first thermal fluid within the thermal fluid tank **126** has attained a first predetermined temperature or temperature range. Upon the first thermal fluid being heated to the first predetermined temperature, the controller **182** can generate a signal used to operate the clutch **164** in a disengaged state such that the force generated by the driver **128** is no longer transmitted to the second drive shaft **166**. For example, as discussed above, according to the exemplary embodiment, the controller **182** can generate a signal for electrical power be delivered to the coil **170** so as to generate a magnetic force that can overcome the biasing force of the biasing element **174**, thereby facilitating the clutch body **172** being displaced to the second, disengaged position. With the clutch body **172** at the second, disengaged position, rotation of the second drive shaft **166** can cease, thereby deactivating operation of the second fluid pump **142**.

(55) With respect to the first thermal fluid, the standpipe **138** can be used to draw first thermal fluid from an upper region of the thermal fluid tank **126** into the first fluid pump **124**. As previously discussed, the inlet **140** of the standpipe **138** can be positioned at a vertically elevated position within the thermal fluid tank **126** so that first thermal fluid at or around the upper portion **134** of the thermal fluid tank **126** that has a higher or elevated temperature relative to first thermal fluid located at the lower portion of the thermal fluid tank **126** can be both drawn into the standpipe **138**, and be outputted from the thermal fluid tank **126** via operation of the first fluid pump **124**. The first fluid pump **124** can pump the first thermal fluid out of the thermal fluid tank **126** and through the inlet control valve **120** before the first fluid pump **124** is circulated through the battery **108**. To the extent the first thermal fluid has been heated or warmed, at this stage of the operation heat within the first thermal fluid can be used to warm the battery, including thermal fluid contained within the battery **108**. However, as previously discussed, while the foregoing is discussed in terms of heating

the first thermal fluid and, to the extent necessary, the thermal fluid within the battery, according to certain embodiments, once either or both the first thermal fluid and thermal fluid within the battery reach elevated temperatures, the first thermal fluid can subsequently be involved with the removal of heat from the battery **108**, including removing heat from thermal fluid within the battery **108**.

(56) After being circulated through the battery **108**, the first thermal fluid can flow through the outlet control valve **122** and back into the thermal fluid tank **126**. According to certain embodiments, the first thermal fluid can be returned to the lower portion **136** of the thermal fluid tank **126** such that the returned first thermal fluid has an opportunity to be heated via heat provided by the second thermal fluid passing through the first heat exchanger **148** before being subsequently drawn into the standpipe **138**.

(57) FIG. **2** illustrates a scenario similar to that shown in FIG. **1**, but in which the first thermal fluid is circulated about the bypass flow path **116a-2** of the first fluid circuit **116a**. Thus, in the scenario shown in FIG. **2**, the second fluid pump is again being operated such that the second thermal fluid flows through the second fluid circuit **116b** in a manner similar to that discussed above with respect to FIG. **1**. Further, the first fluid pump **124** is also operated such that first thermal fluid is drawn into the standpipe **138** and is outputted from the thermal fluid tank **126**. However, unlike the scenario shown in FIG. **1**, in FIG. **2** the inlet and outlet control valves **120**, **122** have, in response to a signal provided by the controller **182**, been placed at the second position at which the bypass flow path **116a-2** is open, and the primary flow path **116a-1** is closed. As previously discussed, with the bypass flow path **116a-2** open, rather than first thermal fluid flowing through the battery **108** as the first thermal fluid is circulated through the first fluid circuit **116a**, the first thermal fluid instead flows through the bypass conduit **118**, thereby bypassing the battery **108**.

(58) The bypass flow path **116a-2** can be used in an attempt to quicken the time to heat the first thermal fluid. For example, as the first thermal fluid does not flow through the battery **108**, to the extent the first thermal fluid has been warmed or otherwise contains heat, while the bypass flow path **116a-2** is open, the heat in the first thermal fluid is not transferred to thermal fluid within the battery **108**. Instead, the first thermal fluid outputted from the thermal fluid tank **126** is returned to the thermal fluid tank **126**. Heat within the first thermal fluid can then be used to at least assist in heating the first thermal fluid within the thermal fluid tank **126**. Accordingly, to the extent there is heat within the first thermal fluid, by using the bypass flow path **116a-2**, the heat can generally be retained within the first thermal fluid, which can assist in decreasing the time before the first thermal fluid reaches an elevated temperature.

(59) Whether the bypass flow path **116a-2** is, or is not, to be opened, can be based on a variety of criteria. For example, according to certain embodiments, the temperature sensor **188** within the thermal fluid tank **126** can provide information to the controller **182** indicating the first thermal fluid is at, or below, a second predetermined temperature such that the extent the temperature of the first thermal fluid is to be raised warrants at least temporary use of the bypass flow path **116a-2**. According to such an embodiment, upon the temperature of the first thermal fluid satisfying, or exceeding, the second predetermined temperature, the controller **182** can issue a signal to adjust the inlet and outlet control valves **120**, **122** from being at the second position to the first position so as to close the bypass flow path **116a-2** and open the primary flow path **116a-1**.

(60) Additionally, or alternatively, according to certain embodiments, the bypass flow path **116a-2** can be open when the controller **182** receives information indicating the second or third thermal fluids are at, or below, a third predetermined temperature that may indicate the second or third thermal fluids currently lack sufficient heat for heating the first thermal fluid. The third predetermined temperature may be the same as, or different than, the second predetermined temperature utilized in evaluating the temperature of the first thermal fluid. According to such an embodiment, upon the temperature of the second or third thermal fluid satisfying, or exceeding, the third predetermined temperature, such as reaching a temperature that can relatively quickly heat the first thermal fluid, the controller **182** may issue a signal to the inlet and outlet control valves **120**,

122 that closes the bypass flow path **116a-2** and opens the primary flow path **116a-1**.

(61) Regardless of whether the first thermal fluid is being circulated through the bypass flow path **116a-2** or the primary flow path **116a-1**, while the first thermal fluid is being heated or warmed, the clutch **164** can remain engaged such that the force provided by the driver **128** is operating both the first fluid pump **124** and the second fluid pump **142**. Moreover, as discussed above, the second fluid pump **142** can continue to be operated at least until the first thermal fluid reaches a temperature that satisfies or exceeds the first predetermined temperature. As previously discussed, upon the controller **182** determining the first thermal fluid satisfies or exceeds the first predetermined temperature, the controller **182** can issue a signal for the disengagement of the clutch **164** such that the second drive shaft **166** is not rotated using power from the driver **128**, and thereby deactivates the second fluid pump **142**.

(62) While FIGS. **1** and **2** are discussed above with respect to using the second thermal fluid of the second fluid circuit **116b** to heat the first thermal fluid, the third thermal fluid of the third fluid circuit **116c** can be used in a similar matter for heating the first thermal fluid. For example, FIG. **3** illustrates a scenario in which the circuit control valve **158** has been, via a signal provided by the controller **182**, placed at a second position. With the circuit control valve **158** at the second position, a third fluid pump **157** of the external device **112** or external system **104**, including, for example, a water pump, can be utilized to circulate third thermal fluid from the pressurized reservoir **156** and through the third fluid circuit **116c**. Such a scenario can arise for a variety of reasons. For example, such a scenario can arise in response to the controller **182** determining the battery **108** lacks sufficient available electrical power to supply electrical power for the operation of the electric heater **146**, or lacks sufficient electrical power to accommodate operation of the electric heater **146** while also providing electrical power for other operations of the thermal management system **100** or associated electric vehicle **106**. According to certain embodiments, in such a situation, the controller **182**, including the processor **184**, can output a signal to inform an operator of electric vehicle **106** that the battery **108** will not, or cannot, supply electrical power to the battery **108**. Such a signal for communicating to the operator the status of the battery **108** can be provided to an information device or input/output device **192**, such as, for example, a monitor, display, touch screen, speaker, or light that may be positioned in the operator cab **110** or on a mobile device, smartphone, or tablet of the operator. Moreover, such a signal can inform the operator to connect the inlet and outlet conduits **118b**, **118c** to the associated inlet and outlet port **160a-b**, **162a-b** so as to fluidly couple the primary and external systems **102**, **104**, and, moreover, to complete the third fluid circuit **116c**. According to certain embodiments, the controller **182** can await a signal from the sensor **190** indicating the portion **116c-2** of the third thermal fluid circuit **116c** residing with the external device **112** is fluidly coupled to the portion **116c-1** of the third thermal fluid circuit **116c** residing with the primary system **102** before operating the circuit control valve **158** and attempting to draw third thermal fluid from the pressurized reservoir **156** into the primary system **102**, and, moreover, into the second portion **116c-2** of the third fluid circuit **116**. Alternatively, the controller **182** can await receipt of a signal inputted by the operator prior to adjusting the circuit control valve **158** and attempting to draw third thermal fluid from the pressurized reservoir **156** of the third fluid circuit **116c** into the primary system **102**.

(63) With the external device **112** fluidly coupled to the primary system **102**, the circuit control valve **158** the driver **128** can provide a force that operates the first fluid pump **124** while the third fluid pump **157** is used to circulate the third thermal fluid. As previously mentioned, in such a situation, the first thermal fluid can be circulated in a manner similar to that discussed above with respect to either FIG. **1** or **2**. Further, as previously discussed, in such scenarios, the operation of the third fluid pump **157** can draw heated third thermal fluid from the pressurized reservoir **156** and into the first heat exchanger **148** in the thermal fluid tank **126**. At the first heat exchanger **148**, heat entrained in the third thermal fluid can be transferred to at least a portion of the first thermal fluid without the third thermal fluid being mixed with the first thermal fluid. The at least partially cooled

third thermal fluid can be circulated from the first heat exchanger **148**, out of the thermal fluid tank **126**, and into the second heat exchanger **152**. At least a portion of the heat remaining within the third thermal fluid can be transferred to air that is to be circulated in the operator cab **110**, thereby at least assisting in heating the operator cab **110**. As previously discussed, such a heat exchange at the second heat exchanger **152**, and heating of the operator cab **110**, can be assisted by the operation of the fan **154**, the cooled third thermal fluid can then be circulated to the heat exchanger **114** of the external device **112** before being returned to the pressurized reservoir **156**. Such a process can continue until the controller **182** determines the first thermal fluid satisfies or exceeds the first predetermined temperature. Following a determination that the first thermal fluid satisfies or exceeds the first predetermined temperature, the controller **182** can issue a signal for the detachment of the inlet and outlet conduits **118b**, **118c** from the inlet and outlet ports **160a-b**, **162a-b**, respectively. Additionally, or optionally, the circuit control valve **158** can be operated to return to the first position.

(64) Referring to FIGS. **1-3**, as previously, mentioned, when the electric vehicle **106** is expected to sit outside in cold weather without charging of the battery **108**, the electric vehicle **106** can also benefit from pumping or purging the first thermal fluid from the battery **108** and storing the first thermal fluid in the insulated thermal fluid tank **126** overnight. Such a condition can allow for more efficient warming of the first thermal fluid overnight, while minimizing the loss of heat in the battery **108** and associated conduits **118** of the first fluid circuit **116a**. Thus, according to certain embodiments, the primary fluid circuit **116** can include an air tube **194** that can provide a supply of air that is to be circulated through at least the battery **108** and associated conduits **118** of the primary fluid circuit **116a** in connection with the purging of the first thermal fluid to the thermal fluid tank **126**. According to certain embodiments, the inlet of the air tube **194** can be positioned within the thermal fluid tank **126** at a location above an anticipated vertical height of the first thermal fluid within the tank **126**. The inlet of the air tube **194** can, however, be positioned at a variety of other locations, including outside of the thermal fluid tank **126**. A purge control valve **196** can be operably positioned to control whether first thermal fluid is to be drawn through the standpipe **138** and into the first fluid pump **124**, or if the first fluid pump **124** is instead to receive air that flows into the air tube **194**. According to the illustrated embodiment, operation of the purge control valve **196** can be controlled by a signal received from the controller **182**.

(65) Whether the purge control valve **196** is to be changed from being at a first position in which first thermal fluid is supplied to the first fluid pump **124**, or at a second position at which air is supplied to the first thermal pump can be based on a variety of factors. For example, the controller **182** can be configured to determine if the electric vehicle **106** has not been operated for a predetermined time period, and that the battery **108** has not been charged during that time period. If the controller **182** can also be configured to determine whether ambient temperatures in the environment are at, or below, a predetermined temperature. The controller **182** determines one or more, if not all, of such conditions exist, among other conditions, the controller **182** can be configured to automatically generate a signal to place the purge control valve **196** at the second position, and to activate the driver **128** such that the first fluid pump **124** circulates air into the first fluid circuit **116a** so as to displace at least a portion, if not all, of the first thermal fluid in the battery **108** and associated conduits **118** of the first fluid circuit **116a** into the thermal fluid tank **126**. Alternatively, or additionally, the controller **182** can be adapted to initiate such placement of the purge control valve **196** at the second position and operation of the driver **128** in response to the controller receiving a purge command from the operator.

(66) According to certain embodiments, the controller **182** can be configured to operate the driver **128** for such purging of the first thermal fluid into the thermal fluid tank **126** for a predetermined period of time. The predetermined period of time may be based on the estimated time needed for a certain portion, including most or all, of the first thermal fluid to be purged into the thermal fluid tank **126**. Additionally, or alternatively, the controller **182** can receive information from one or

more sensors indicating the purging is complete. For example, according to certain embodiments, one or more sensors can indicate a change in flow rate or mass of the first thermal fluid that can indicate the first thermal fluid has been purged into the thermal fluid tank **126**. Upon completion of the purging operation, the controller **182** can issue a signal to place the purge control valve **196** back at the first position, and to deactivate operation of the driver **128**.

(67) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A thermal management system comprising: a first fluid pump and a second fluid pump, the first fluid pump positioned for at least partial submersion in a first thermal fluid in a thermal fluid tank of a first thermal circuit, the second fluid pump positioned for at least partial submersion in a second thermal fluid in a first pressurized reservoir of a second fluid circuit; a first drive shaft coupled to a driver and the first fluid pump, the first fluid pump being operated by a rotational displacement of the first drive shaft by the driver, the first fluid pump being configured to provide a first force to circulate the first thermal fluid through the first fluid circuit, a second portion of the first drive shaft positioned in an interior area of the first thermal fluid tank, a first portion of the first drive shaft being positioned outside of the thermal fluid tank and not directly physically attached to the second portion of the first drive shaft; a second drive shaft coupled to the second fluid pump and coupled to the first portion of the first drive shaft by a clutch, the second fluid pump being operated by a rotational displacement of the second drive shaft to provide a second force to circulate a second thermal fluid at least through a heat exchanger of the second fluid circuit positioned in the thermal fluid tank; and a magnetic coupling having a first portion and a second portion, the first portion of the magnetic coupling attached to the first portion of the first drive shaft and positioned outside of the thermal fluid tank, the second portion of the magnetic coupling coupled to the second portion of the first drive shaft and positioned in the interior area of the thermal fluid tank; wherein the magnetic coupling transfers a force between the first and second portions of the magnetic coupling to rotatably displace the first portion of the magnetic coupling, and thereby rotate the second portion of the second drive shaft, with the rotational displacement of the first portion of the magnetic coupling as the first portion of the first drive shaft is rotably displaced by operation of the driver, and wherein the clutch is selectively displaceable from an engaged position at which the rotational displacement of the first portion of the first drive shaft is transmitted by the clutch to facilitate the rotational displacement of the second drive shaft, and a disengaged position at which the rotational displacement of the first portion of the first drive shaft is not transmitted by the clutch to facilitate the rotational displacement of the second drive shaft.
2. The thermal management system of claim 1, wherein the first drive shaft does pass through or breach the thermal fluid tank.
3. The thermal management system of claim 1, further comprising a memory device coupled with at least one processor, the memory device including instructions that when executed by the at least one processor cause the at least one processor to: (a) determine if a temperature of the first thermal fluid in the thermal fluid tank satisfies a predetermined threshold; (b) generate, in response to the temperature being determined to not satisfy the predetermined threshold, a signal to displace the clutch to the engaged position; and (c) generate, in response to the temperature being determined to satisfy the predetermined threshold, a signal to displace the clutch to the disengaged position, wherein step (a) is repeated until the temperature is determined to satisfy the predetermined threshold.
4. The thermal management system of claim 3, wherein the first fluid circuit is configured to

circulate the first thermal fluid to a battery, and wherein the memory device further includes instructions that when executed by the at least one processor cause the at least one processor to: determine if a predetermined time period has expired since the battery has been charged; generate, if the predetermined time period is determined to have expired, a first signal to open a purge control valve to receive air into the first fluid circuit and a second signal to displace the clutch to the disengaged position; and generate, with the purge control valve open, a signal to operate the driver, operation of the driver rotating the first drive shaft to operate the first fluid pump to draw air into the first fluid circuit.

5. The thermal management system of claim 4, wherein the memory device further includes instructions that when executed by the at least one processor cause the at least one processor to determine if an ambient temperature is below a predetermined temperature, and wherein the first and second signals are generated if the predetermined time period has expired and the ambient temperature is below the predetermined temperature.

6. The thermal management system of claim 1, wherein the clutch is an electromagnetic clutch.

7. The thermal management system of claim 1, wherein the first fluid circuit is fluidly isolated from the second fluid circuit.

8. The thermal management system of claim 1, further comprising a second pressurized reservoir, and wherein the second force provided by the second fluid pump draws the second thermal fluid from the second pressurized reservoir.

9. The thermal management system of claim 1, wherein the driver is a single driver.

10. The thermal management system of claim 1, wherein the first fluid circuit is configured to circulate the first thermal fluid at least to a battery.

11. A method for operating a thermal management system, the method comprising: determining if a temperature is below a predetermined temperature; activating, in response to determining the temperature is below the predetermined temperature, a magnetic clutch to magnetically couple a first portion of the magnetic clutch to a second portion of the magnetic clutch, the first portion of the magnetic clutch being coupled to a first portion of a first drive shaft that is positioned outside of a thermal fluid tank, the second portion of the magnetic clutch being coupled to a second portion of a first drive shaft that is positioned in an interior area of the thermal fluid tank, the second portion of the first drive shaft being coupled to a first fluid pump; activating a driver to provide a drive force to rotatably displace the first portion of the first drive shaft, the rotational displacement of the first portion of the first drive shaft being transmitted via the activated magnetic clutch to facilitate rotational displacement of the second portion of first drive shaft, an operation of the first fluid pump being driven by rotational displacement of the second portion of first drive shaft; engaging a clutch that is coupled to the first portion of the first drive shaft and a second drive shaft, the clutch configured to transmit a portion of the drive force to rotatably displaced the second drive shaft, an operation of a second fluid pump of a first pressurized reservoir being driven by rotational displacement of the second drive shaft; circulating, via the operation of the first fluid pump, at least a portion of a first thermal fluid out of the thermal fluid tank and along at least a portion of a first fluid circuit; drawing, via the operation of the second fluid pump, a second thermal fluid into a heat exchanger in the thermal fluid tank; determining if the temperature has increased to at least satisfy the predetermined temperature; and deactivating, in response to the temperature satisfied the predetermined temperature, the operation of the second fluid pump by disengaging the clutch.

12. The method of claim 11, wherein the temperature is a temperature of a first thermal fluid in a thermal fluid tank.

13. The method of claim 12, wherein the first fluid pump is at least partially submerged in the first thermal fluid in the thermal fluid tank.

14. The method of claim 13, wherein the first drive shaft does not pass through or breach the thermal fluid tank.

15. The method of claim 11, wherein the first fluid circuit is configured to circulate the first thermal

fluid to a battery, and wherein the method further comprises: determining if a predetermined time period has expired since the battery has been charged; generating, if the predetermined time period is determined to have expired, a first signal to open a purge control valve to receive air into the first fluid circuit and a second signal to displace the clutch to a disengaged position; and generating, with the purge control valve open, a signal to operate the first fluid pump to draw air into the first fluid circuit.

16. The method of claim 15, further comprising determining if an ambient temperature is below a predetermined temperature, and wherein the first and second signals are generated if the predetermined time period has expired and the ambient temperature is below the predetermined temperature.

17. The method of claim 11, wherein the first fluid circuit is configured to circulate the first thermal fluid to a battery, and wherein the method further comprises: generating, in response to the temperature being determined to at least satisfy the predetermined temperature, a signal for the battery to either (1) deliver an electrical power the clutch facilitates displacement of the clutch an engaged position, or (2) cease delivery of the electrical power to the clutch to facilitate the displacement of the clutch to a disengaged position.

18. The method of claim 11, further comprising drawing the second thermal fluid into the second fluid pump from the first pressurized reservoir.

19. The method of claim 11, further comprising continuously operating the first fluid pump, but not the second fluid pump, while the driver is activated.

20. The method of claim 11, further comprising drawing the first thermal fluid into the first fluid pump through a standpipe in the thermal fluid tank.
